Javad Maheri

+33 6 44 98 96 50 | j.maheri95@gmail.com | Sophia Antipolis, France
linkedin.com/in/javad-maheri | github.com/JavadMaheri | scholar.google.com/JavadMaheri

Profile

Ph.D. in machine learning and distributed systems (defended July 2026, Sorbonne Université / EURECOM), **available immediately**. I design learning and computing systems that are provably efficient and that run in practice: matching lower bounds paired with explicit constructions, validated in PyTorch on multi-GPU clusters. Hands-on practical experience across machine learning, deep and graph neural networks for industrial safety monitoring, large-scale graph ranking on Spark, and generalization theory.

Technical Skills

Languages: Python (advanced), MATLAB, C/C++, LaTeX; SQL
ML / DL: PyTorch, TensorFlow, NumPy, SciPy, scikit-learn; graph neural networks, transformers and attention, CNNs, transfer learning
Distributed & Data: Apache Spark / GraphX, multi-GPU training, communication-efficient training, MapReduce-style workloads, cluster experiment pipelines
Developer Tools: Git, Linux, PyCharm, Jupyter, mixed-precision training, reproducible multi-run evaluation
Research Areas: Generalization bounds (PAC-Bayes, chaining mutual information), information theory, high-dimensional probability, spectral graph theory, graph signal processing, convex optimization, combinatorial design
Languages (spoken): English (fluent), French

Experience

Research Scientist, Distributed Learning & Computing Apr. 2023 – July 2026
EURECOM / Sorbonne Université | Huawei France Chair for Future Wireless Networks *Biot, France*

- Built the first **universally optimal** allocation schemes for a general class of distributed problems: one task-blind assignment that is simultaneously optimal across all downstream workloads, removing the need to re-plan data placement per job.
- Derived matching information-theoretic lower bounds and **explicit combinatorial constructions** achieving them, cutting inter-node communication load to its provable minimum on MapReduce-style workloads.
- Designed and ran the full experimental program in Python (PyTorch, NumPy) .

Machine Learning Engineer – GNNs for HSE Video Analytics April 2021 – Jan. 2022
Industrial Power Generation Facility (MAPNA) *[Tehran]*

- Built a **scene-graph GNN** on top of a detection backbone – people, machinery, PPE and restricted zones as nodes, spatial and temporal relations as edges – classifying Health, Safety & Environment violations from *interactions* (unsafe proximity, missing helmet or vest, restricted-area entry) rather than isolated per-frame detections.
- Cut false alarms substantially against the per-frame detector baseline by letting message passing resolve occlusion and viewpoint ambiguity across overlapping cameras – the failure mode that had made the incumbent system unusable for operators.
- Formulated **camera configuration as graph optimization**: modelled the installation as a coverage graph over field-of-view overlaps and hazard zones, then ranked placement, orientation and resolution settings to improve hazard-zone coverage without adding hardware.

Machine Learning Engineer – Graph Ranking & Structure Learning March 2022 – Dec. 2022
[Zarebin] *Tehran*

- Implemented **PageRank and personalized PageRank** over multi-million-edge graphs on **Apache Spark / GraphX**, using the Pregel-style message-passing API to keep iterative computation distributed rather than collapsing it onto a driver node.
- Learned **graph structure from data** instead of assuming it: convex graph Laplacian estimation under smoothness and sparsity constraints to infer market-dependency graphs from raw price series.
- Benchmarked learned graphs against correlation baselines on downstream ranking and clustering quality; delivered the pipeline to the client's analytics team.

Teaching Assistant – Information Theory II, Probability & Statistics, DSP, GSP 2019 – 2023
EURECOM and Sharif University of Technology *France*

- Ran tutorials, built problem sets and lab exercises, and mentored students across five courses spanning theory and implementation.

Projects

CassavaNet | *PyTorch, CNNs, Transfer Learning, Mixed Precision* Kaggle Competition

- Five-class crop-disease recognition on a ∼21k-image dataset where the difficulty is the data, not the architecture: severe class imbalance, noisy crowd-sourced labels, uncontrolled field photography.
- Built the pipeline end to end – stratified cross-validation, augmentation policy (crop, flip, colour jitter, cutout-style occlusion), fine-tuning of pretrained CNN backbones, test-time augmentation at inference.
- Handled label noise and imbalance explicitly through loss reweighting and label smoothing, rather than chasing leaderboard position with ensembles that would not survive deployment.
- Optimized for **accuracy per unit of compute**: mixed-precision training, LR scheduling and early stopping, with backbone size chosen from the accuracy-versus-cost curve; diagnosed residual errors by class to separate genuine visual ambiguity from mislabelled ground truth.

ChainMI | *Information Theory, PAC-Bayes, High-Dimensional Probability* Manuscript in preparation for ICML

- Applied chaining-based mutual-information techniques to sharpen PAC-Bayes and single-sample generalization bounds for stochastic learning algorithms.
- Resulting estimates track generalization in overparameterized regimes where classical uniform-convergence bounds are vacuous.

Attention Mechanism | *PyTorch, Transformers* Independent study

- Implemented scaled dot-product and multi-head self-attention from scratch without framework shortcuts; analyzed expressivity, scaling behavior and approximation properties relevant to language modelling.

LaplaceFit | *Convex Optimization, Spectral Graph Theory* Research project

- Formulated and solved convex programs for learning graph Laplacians in bipartite and block-structured graphs; built efficient solvers and characterized their convergence.

CardioSep & NucleoSeg | *Signal Processing, Deep Learning, Medical Imaging* Sharif University

- Adaptive filtering, Kalman filtering and blind source separation to strip ECG artefacts from surface EMG recordings; colozr-transform plus deep-network pipeline for leukocyte nuclei segmentation (published).

Selected Publications

J. Maheri, K. K. Krishnan Namboodiri, P. Elia. *Universal and Asymptotically Optimal Data and Task Allocation in Distributed Computing.* **IEEE Transactions on Information Theory**, accepted 2026.
J. Maheri, K. K. Krishnan Namboodiri, P. Elia. *Order Optimal Task Allocation in Distributed Computing via Interweaved Cliques.* **IEEE ISIT**, 2026.
J. Maheri, P. Elia. *Constructing Hamiltonian Decompositions of Complete k-Uniform Hypergraphs.* **IEEE ISIT**, 2025.
J. Maheri, K. K. Krishnan Namboodiri, P. Elia. *Fundamental Limits of Hypergraph Edge Partitioning under Independent Edge Sampling.* Under review, **SIAM J. Discrete Mathematics**, 2026.
Deriving fundamental limits for **Alistarh**, *Streaming min-max hypergraph partitioning*, **Advances in Neural Information Processing Systems (NeurIPS 2015)**

Awards and Honors

Ranked 7th nationally among 180,000+ candidates, Iranian National University Entrance Examination (2017)
High honors graduate, B.Sc. in Electrical Engineering & Mathematics, Sharif University of Technology (2022)
Fully funded doctoral position, Huawei France Chair for Future Wireless Networks / France 2030 PEPR Networks of the Future (2023–2026)
ISIT 2024 Chess Event: drew against GM Vassilios Kotronias; featured on EURECOM news and the ISIT 2024 website

Education

Sorbonne Université / EURECOM Biot, France
Ph.D. in Communication Systems – defended July 2026 *Apr. 2023 – July 2026*

- Thesis: *Universally Optimal Combinatorial Schemes for Distributed Computing and Learning.* Supervisor: Prof. Petros Elia.

Sharif University of Technology Tehran
B.Sc. in Electrical Engineering & Mathematics (double major), high honors *2017 – 2022*

- Selected grades: Theory of Machine Learning 20/20, Graph Signal Processing 20/20, Biomedical Signal Processing 19.5/20, Information Theory 19.4/20, Deep Learning 19.1/20.